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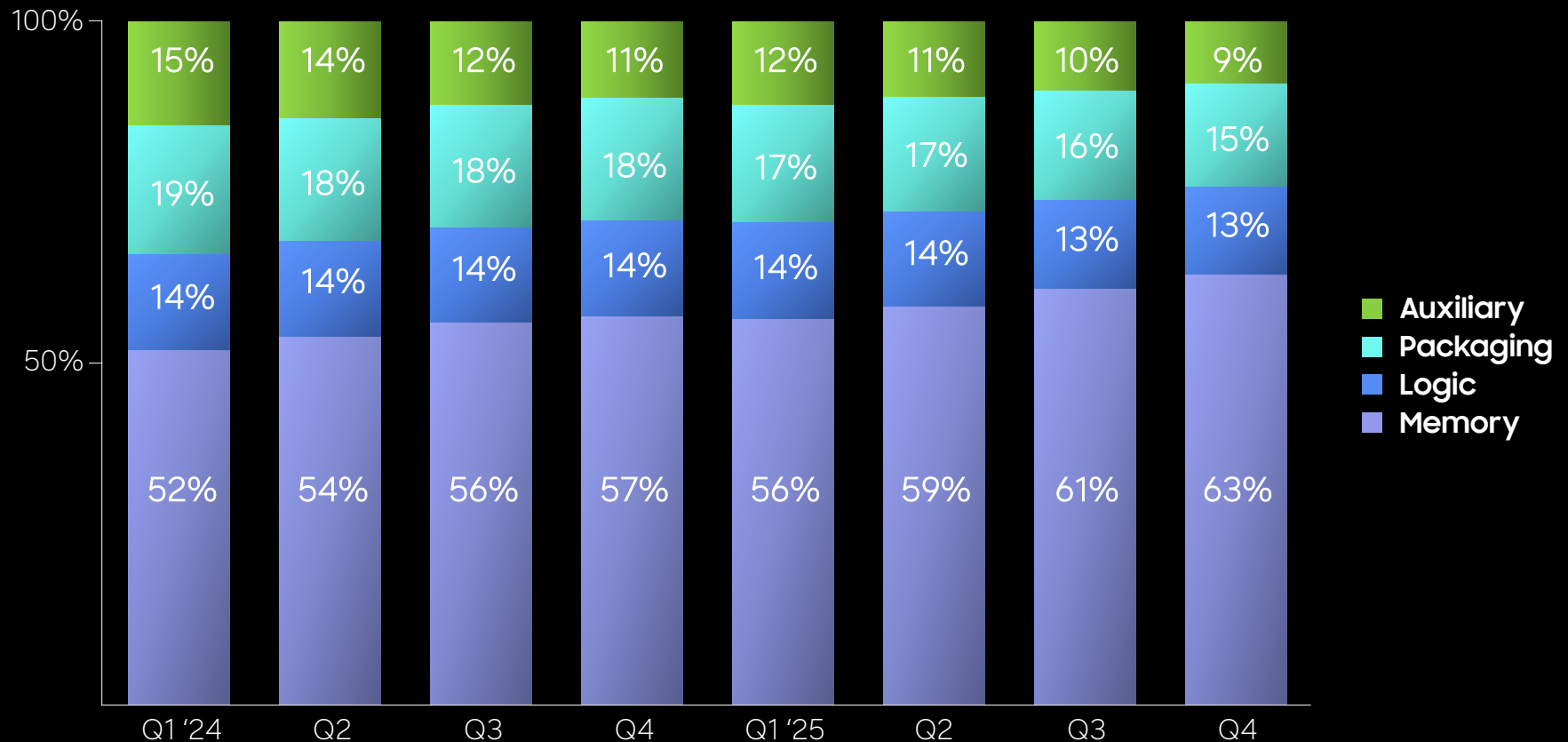
Samsung LPDDR5X-PIM

: World's First LPDDR based Processing-In-Memory (PIM) Solution
for AI Inference

Karam Hwang, Yoonah Paik, Sukhan Lee, Taeyang Jeong, Wontak Han, Byeongho Kim, Sanghoon Cha, Hyeonsu Kim,
Seyoon Kong, Jihyun Choi, Duksung Kim, Doohee Hwang, Jangseok Choi and Kyomin Sohn

AI Chip Component Cost Breakdown

- The Cost of Memory (specifically HBM) takes up the majority of the AI chip component spending
- Cost has grown from 52% (Q1 '24) to 63% (Q4 '25)



V. Somala, "Memory has grown to nearly two-thirds of AI chip component costs", [Epoch AI](#), (accessed Aug. 9, 2026)

The Need for New Memory Solutions

The de facto memory solution to support AI Inference: HBM

- High power consumption
- Complex packaging



To achieve lower cost, various advancements have been made in AI Inference:

- Efficient LLM architecture
- Smaller size models

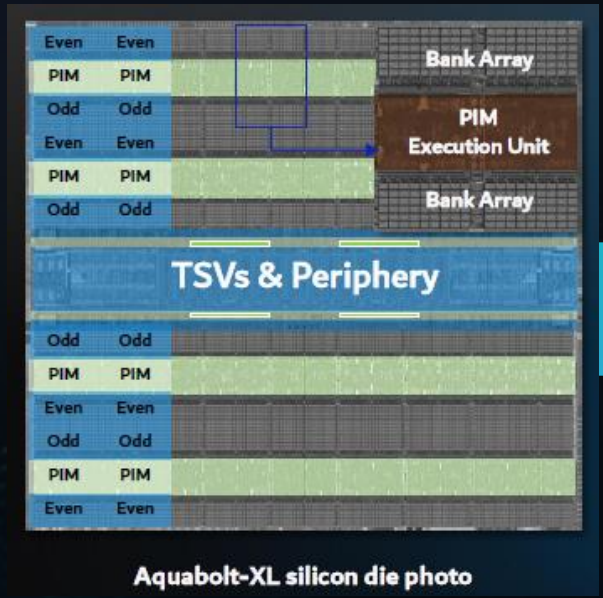


A new memory form factor is required → **LP-PIM**

- Higher bandwidth (compared to normal LPDDR)
- Lower overall power consumption at system level

Progression of Samsung PIM Technology

2021



HBM-PIM Proof of Concept

Aquabolt-XL : Samsung HBM2-PIM with in-memory processing for ML accelerators and beyond

2023

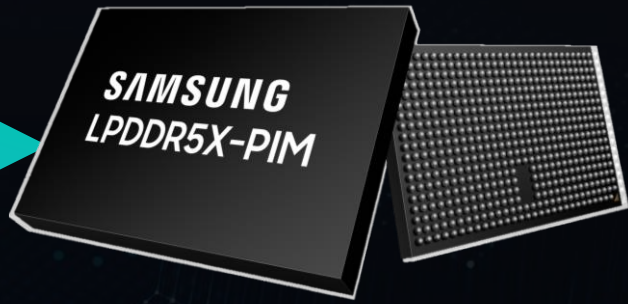
HBM PIM-Cluster



HBM-PIM Cluster Validation & LP-PIM Conceptualization

Samsung PIM/PNM for Transformer based AI

2026

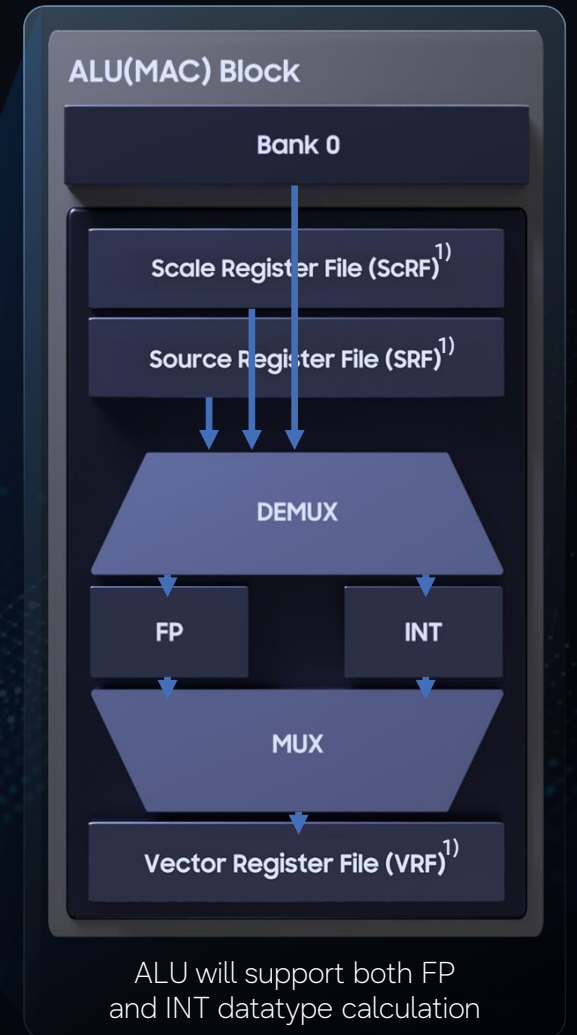
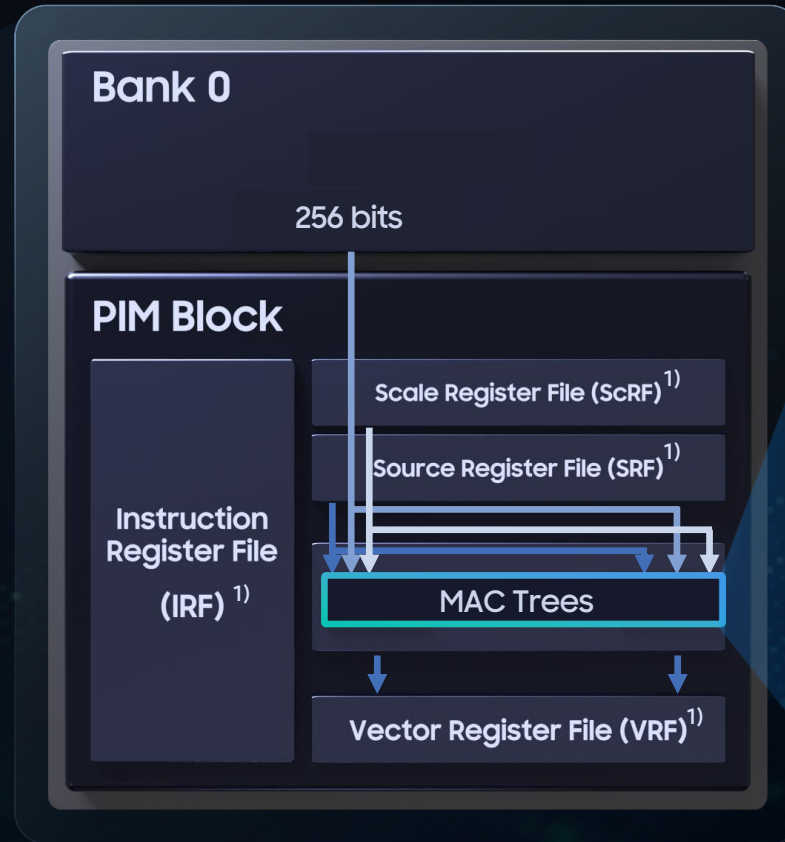


LPDDR5X-PIM Productization

World's First LPDDR-PIM Solution for AI Inference

LPDDR5X-PIM Architecture

- 16 PIM blocks are implemented to all DRAM Banks to maximize internal PIM Bandwidth
- MAC trees operating in parallel



1) Exact description of the terms are provided in Appendix : Terms and Functions
Note : All figures are not to scale

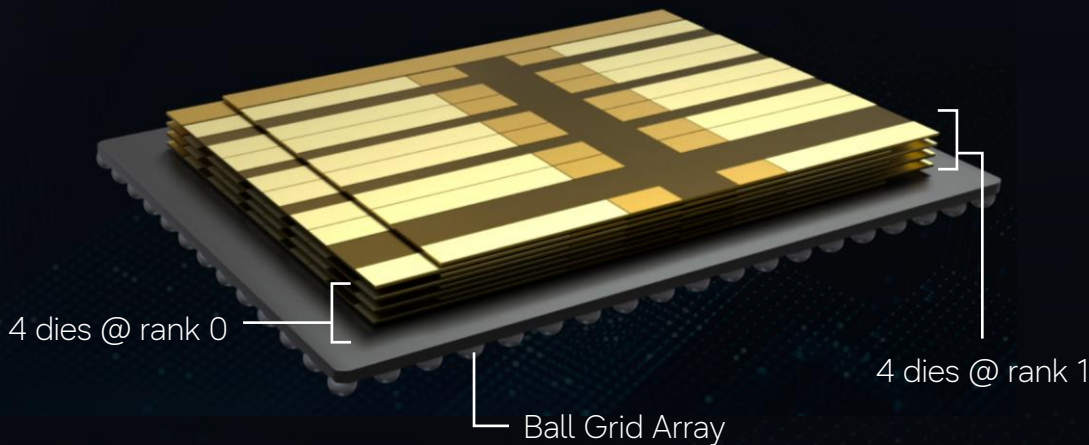
LPDDR5X-PIM Specifications

LPDDR5X-PIM Package

561-ball Package (based on JEDEC standard)

16 GB Density

2 ranks of 4-CH (x64)



Target Application	Server / Mobile / Client
PIM configuration	1 PIM / 1 Bank
DRAM / PIM bandwidth	76.8 GB/s ¹⁾ , 614 GB/s (8x)
Configuration Register ²⁾	256 bit / 1 PIM
Instruction Register File (IRF) ²⁾	1 kbit / 1 PIM
Source Register File (SRF) ²⁾	4 kbit / 1 PIM
Scale Register File (ScRF) ²⁾	2 kbit / 1 PIM
Vector Register File (VRF) ²⁾	1 kbit / 1 PIM

1) Based on a x64 LP5X/LP5X-PIM package operating @ 9600 Mbps frequency

2) Exact description of the terms are provided in Appendix : Terms and Functions

LPDDR5X-PIM Multi-Precision Datatype Support

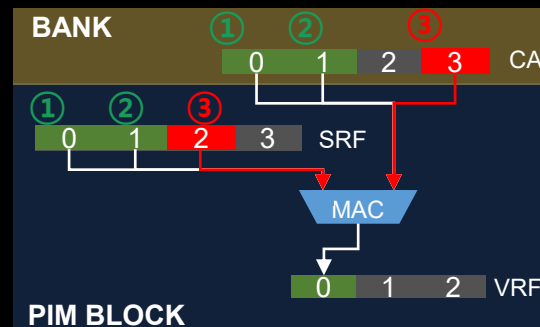
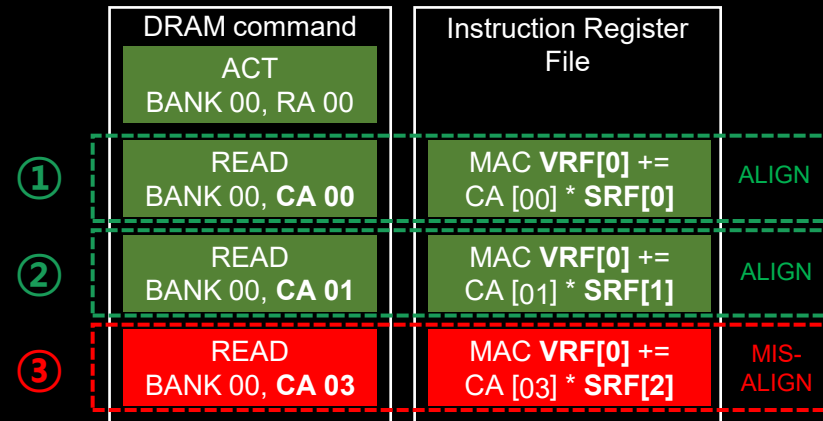
- The first LP-PIM product with multi-precision datatype support
- Can be selected via the 'MAC Precision Datatype Select' and 'MAC Precision Activation Bitwidth' field in the CRF
 - 15 different combinations are supported

Input Data Precision (Activation)	Input Data Precision (Weight)	Computation Power (per PKG)
UINT (8, 16)	SINT4	2.4 TOPs
SINT (4, 8, 16)		
UINT (8, 16)	SINT8	1.2 TOPs
SINT (8, 16)		
FP16	FP8 (E4M3)	1.2 TFLOPs
FP8 (E4M3, E5M2)		
FP16	FP8 (E5M2)	1.2 TFLOPs
FP8 (E4M3, E5M2)		

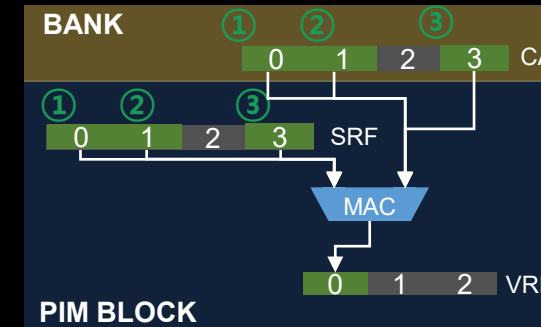
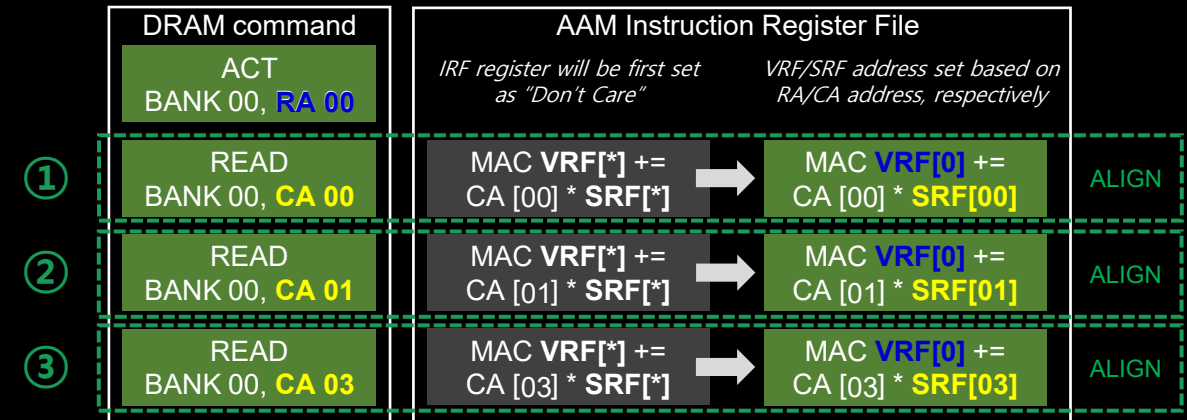
How PIM Utilizes Conventional DRAM Controller

- Utilization of conventional DRAM controller is achieved through Address Align Mode (AAM)
- This novel method leverages the mapping between DRAM addresses and MAC instructions, maintaining re-ordering technique in DRAM controllers to maximize performance

Conventional (non-AAM) Instruction Case

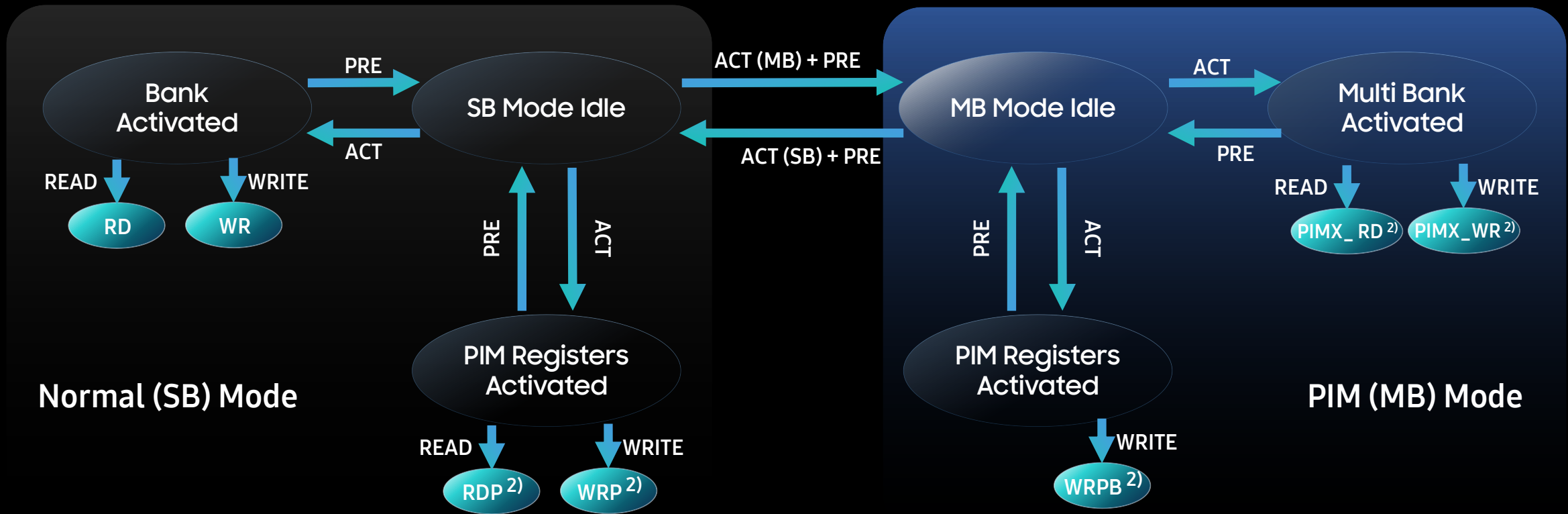


AAM Instruction Case



LPDDR5X-PIM Mode Change

- Requires predefined rows for mode change and PIM register (Defined through MR ¹⁾)
 - Predefined rows for mode change per channel (MB: Multi-bank, SB: Single-bank)
 - A predefined row for register access per bank (PIM Registers)
 - A faster and reliable mode change method compared to previous generation (HBM-PIM)

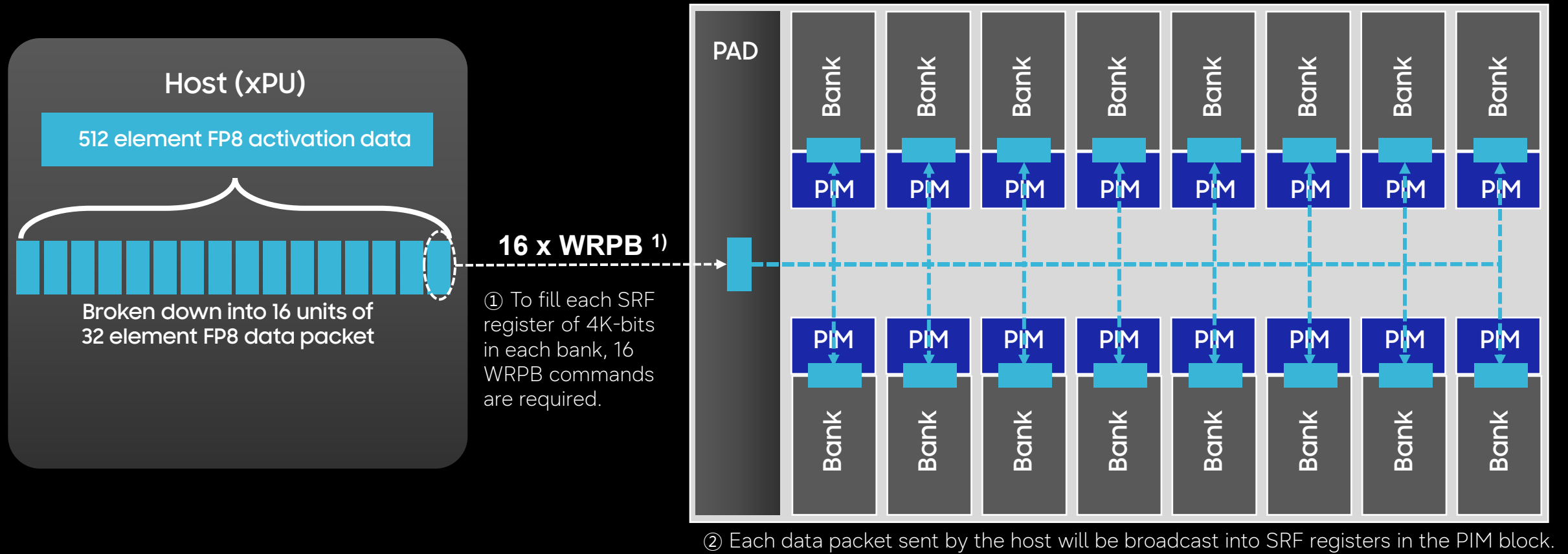


1) MR : Mode Register

2) Exact description of the commands are provided in Appendix : Terms and Functions

LPDDR5X-PIM Operation : Activation Write (WRPB¹⁾)

- The following execution process shows how the PIM will conduct a MAC operation (assumption: weight parameters already written into the cell, currently in PIM MB mode), followed by sending calculated data to host

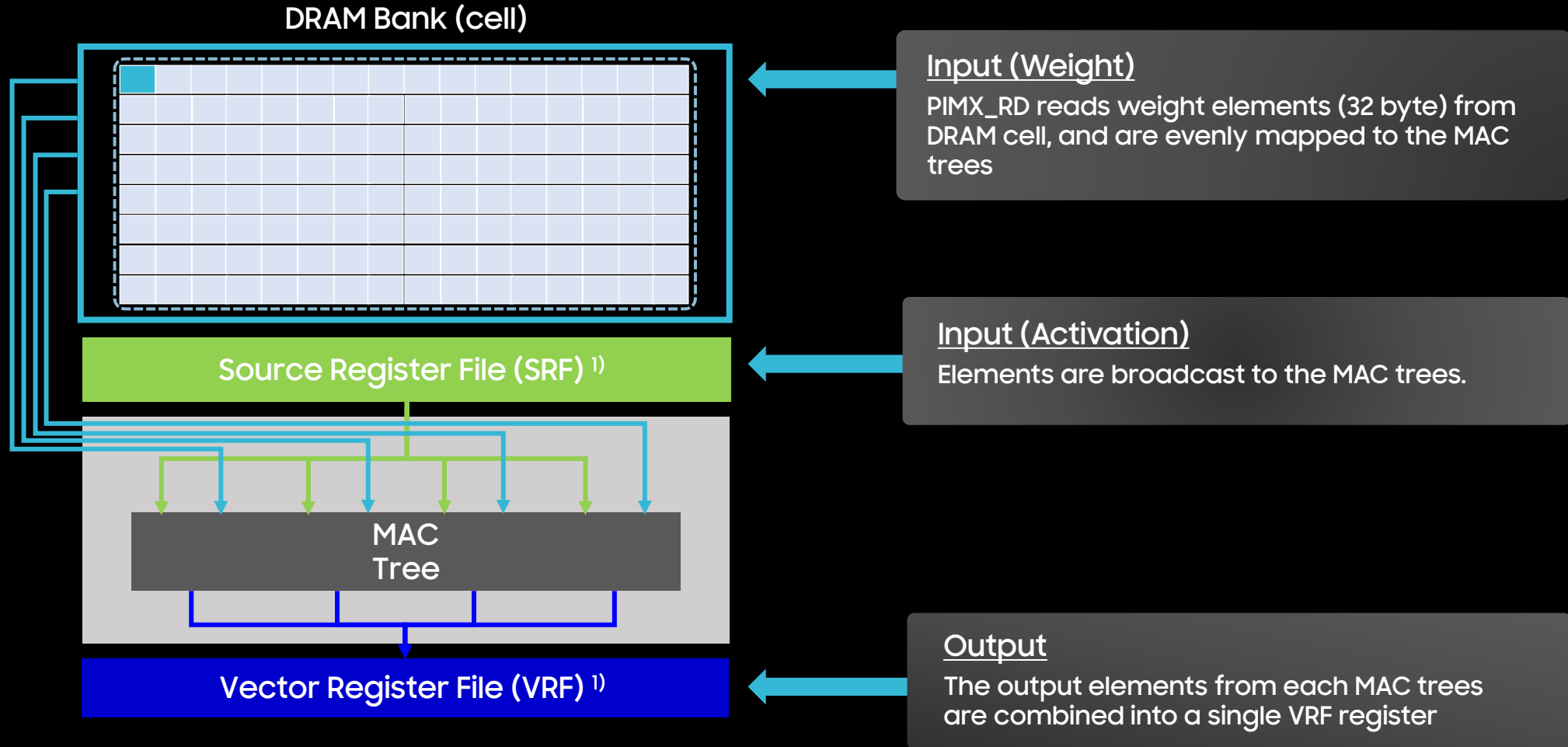


¹⁾ Exact description of the commands are provided in Appendix : Terms and Functions

Note: Example shown depicts 512 elements of FP8 data. Considering activation datatype, number of elements will vary (limited to the 4K-bit SRF register size).

LPDDR5X-PIM Operation : Weight Matrix Load (PIMX_RD¹⁾)

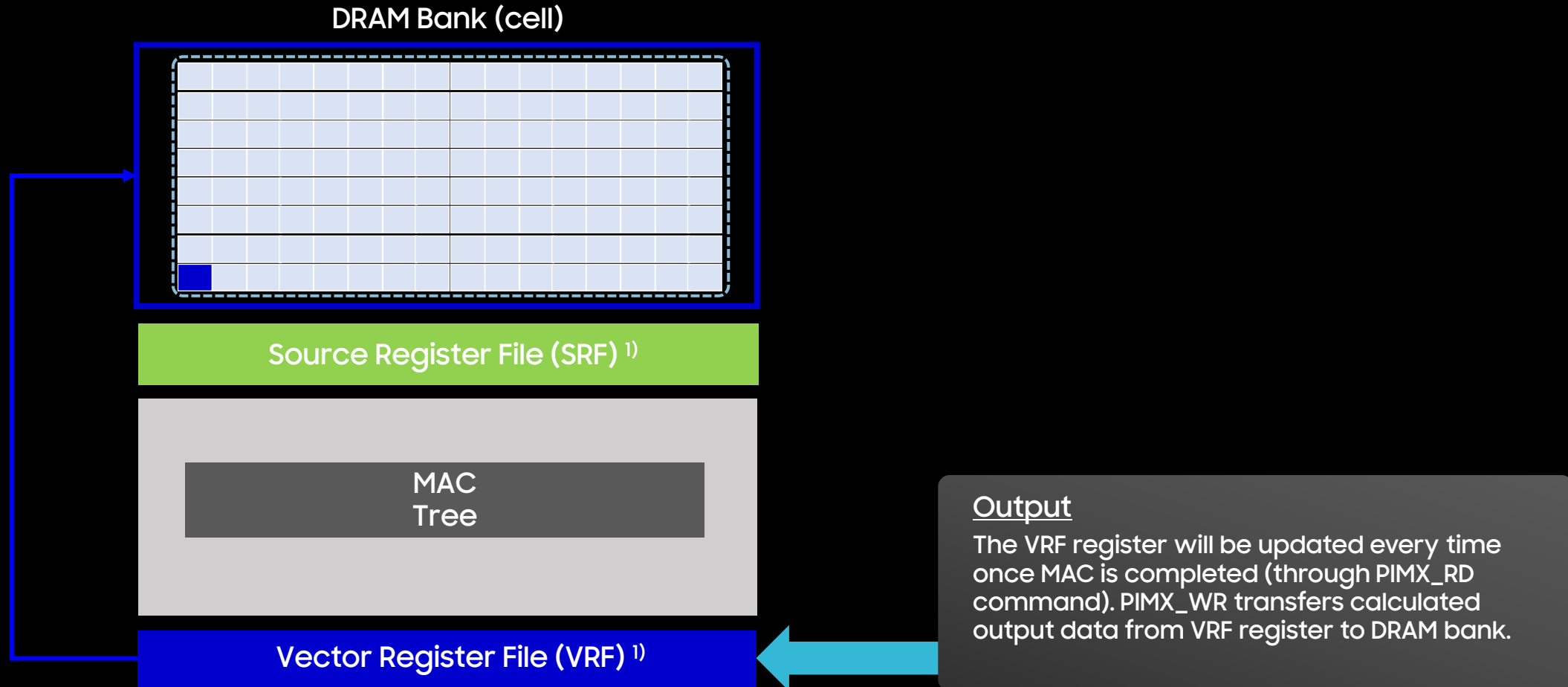
- The following execution process shows how the PIM will conduct a MAC operation (assumption : weight parameters already written into the cell, currently in PIM MB mode), followed by sending calculated data to host



¹⁾ Exact description of the commands are provided in Appendix : Terms and Functions

LPDDR5X-PIM Operation : Vector Store (PIMX_WR¹)

- The following execution process shows how the PIM will conduct a MAC operation (assumption : weight parameters already written into the cell, currently in PIM MB mode), followed by sending calculated data to host

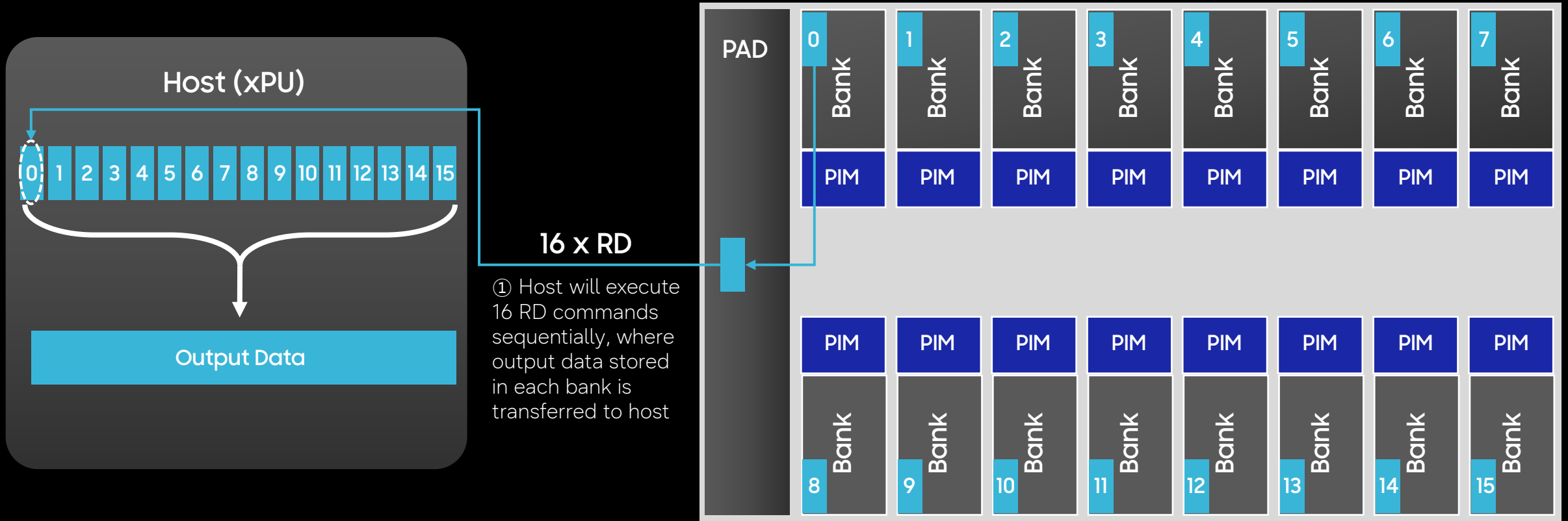


¹⁾ Exact description of the commands are provided in Appendix : Terms and Functions

Note: To move a fully filled VRF register of 1K-bits to the cell, 4 PIMX_WR commands are required. PIMX_RD to PIMX_WR command execution ratio does not have to be 1:1

LPDDR5X-PIM Operation : Vector Read

- From here, the host switches from MB → SB mode
- Here, the host executes 16 RD commands to retrieve calculated PIM data



Note: Example shown depicts 16 RD commands. Based on the size of the VRF register (1K bits), a maximum of 64 RD commands can be executed

Evaluation Setup

- Tested on silicon w/ Samsung Edge AI accelerator SoC (LPDDR5X & LPDDR5X-PIM)
- Model used for evaluation : Llama-3.1-8B
- Context Length : 320 Tokens (Input & Output)
- Applied data type for Activation / Weight / Output : SINT8 / SINT4 / SINT32

Prompt : *"The challenges of building Processing in Memory (PIM) systems?"*

Samsung Edge AI Accelerator SoC w/ LPDDR5X

Processing in Memory (PIM) systems are designed to accelerate data processing and analysis by performing computations directly on data stored in memory. While PIM systems offer significant performance benefits, they also present several challenges, including:

1. **Memory Hierarchy:** PIM systems need to navigate a complex memory hierarchy, which includes cache, main memory, and storage devices. This requires efficient management of memory accesses and data transfer between different levels of the hierarchy.
2. **Memory Bandwidth:** PIM systems require high-bandwidth memory interfaces to move large amounts of data between memory and processing units. This can be challenging, especially when dealing with large datasets.
3. **Scalability:** As PIM systems are designed to handle massive datasets, scalability becomes a significant challenge. PIM systems need to be able to efficiently process data in parallel, without compromising performance or increasing power consumption.
4. **Energy Efficiency:** PIM systems are designed to be energy-efficient, as they operate at the memory level, which is a power-hungry component. PIM systems need to minimize power consumption while maintaining performance.
5. **Memory Error Handling:** PIM systems need to handle memory errors, such as errors caused by faulty memory or data corruption, to ensure reliable operation.
6. **Data Movement:** PIM systems need to efficiently move data between different levels of the memory hierarchy, which can be challenging, especially

TTFT

0.51 s

Output TPS

27.0

Tokens

272

Samsung Edge AI Accelerator SoC w/ LPDDR5X-PIM

Llama-v3-8B-Instruct @4bit weight

The challenges of building PIM(Processing in Memory) systems?

Building Processing in Memory (PIM) systems poses several challenges, including:

1. **Scalability:** PIM systems require a scalable architecture and data management capabilities to handle large volumes of data.
2. **Performance:** PIM systems must handle data processing and storage capacity, memory, and speed. The performance of these systems depends on the processing capacity of the memory.
3. **Data Management:** Managing PIM systems requires a proper data management strategy to ensure the data storage is efficient and the PIM system can efficiently process large amounts of data, and also to ensure store large amounts of data data.
4. **Data Ingestion:** PIM systems should provide data ingestion and processing and also provide integration to other PIM systems.
5. **Data processing speed:** PIM systems must be fast and processing data storage in a few seconds.

TTFT

0.53 s

Output TPS

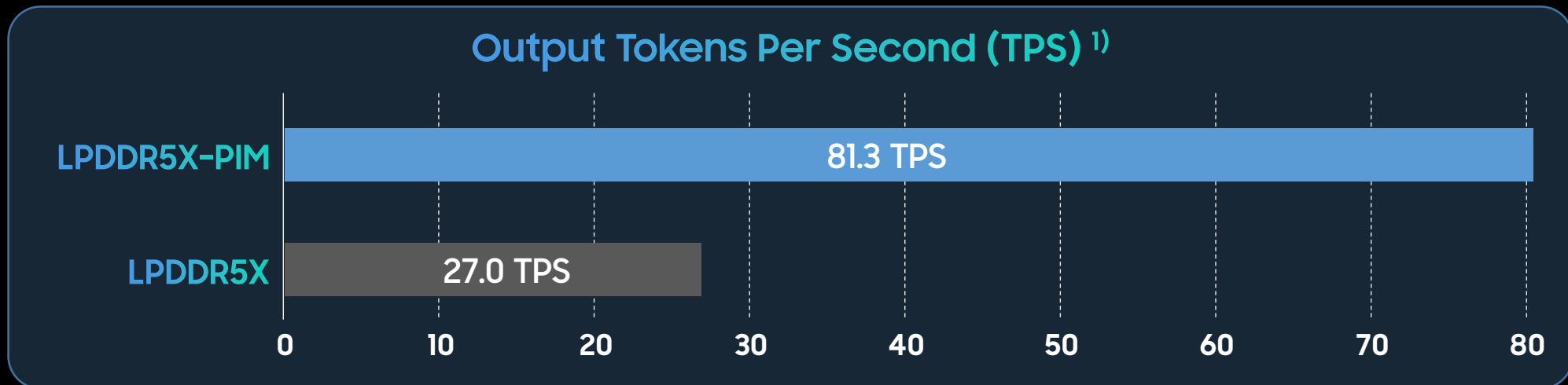
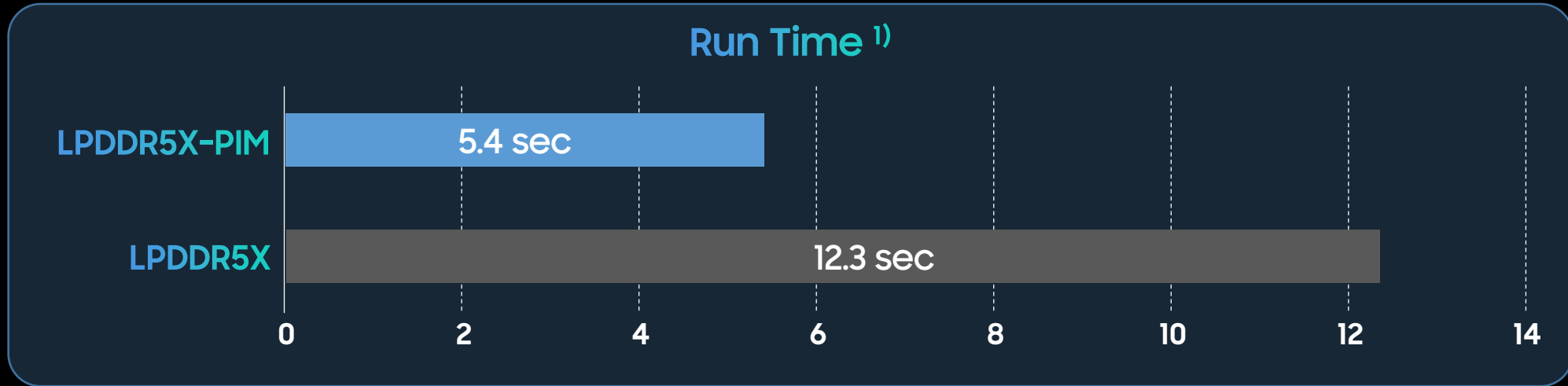
81.3

Tokens

184

Evaluation Results

- Samsung edge AI accelerator SoC performance comparison (LPDDR5X-PIM, LPDDR5X)



1) Performance optimization with PIM is still ongoing

Current Progress in LP-PIM

LPDDR5X-PIM

- Simulator, and datasheet are available upon request
 - The simulator is part of our Software Development Kit (SDK) bundle, which can be referred in the appendix
 - SDK is for reference guidance only, as system operation will vary by host
- More information regarding the simulator can be referred to the following paper : "S. Cha, J. Choi, B. Kim, Y. Paik, S. Lee, and K. Sohn, "LP5X-PIM Sim: A High-Fidelity HW/SW Integrated Simulator for LPDDR5X-PIM," [arXiv:2606.00636](https://arxiv.org/abs/2606.00636), May 2026"

LPDDR6-PIM

- JEDEC 1st full version spec near finalization (specific for mobile/client profile)
- Next JEDEC version is also being considered

Key Takeaways

World's first LPDDR5X-PIM product

- 8× bandwidth compared to conventional LPDDR5X
- Multi-Precision datatype support
- Acceptable for existing SoCs: Drop-in replacement enabled by AAM

LPDDR5X-PIM evaluation results

- 3.01x performance gain over conventional LPDDR5X solution

LPDDR6-PIM

- JEDEC LP6-PIM standardization under progress

THANK YOU

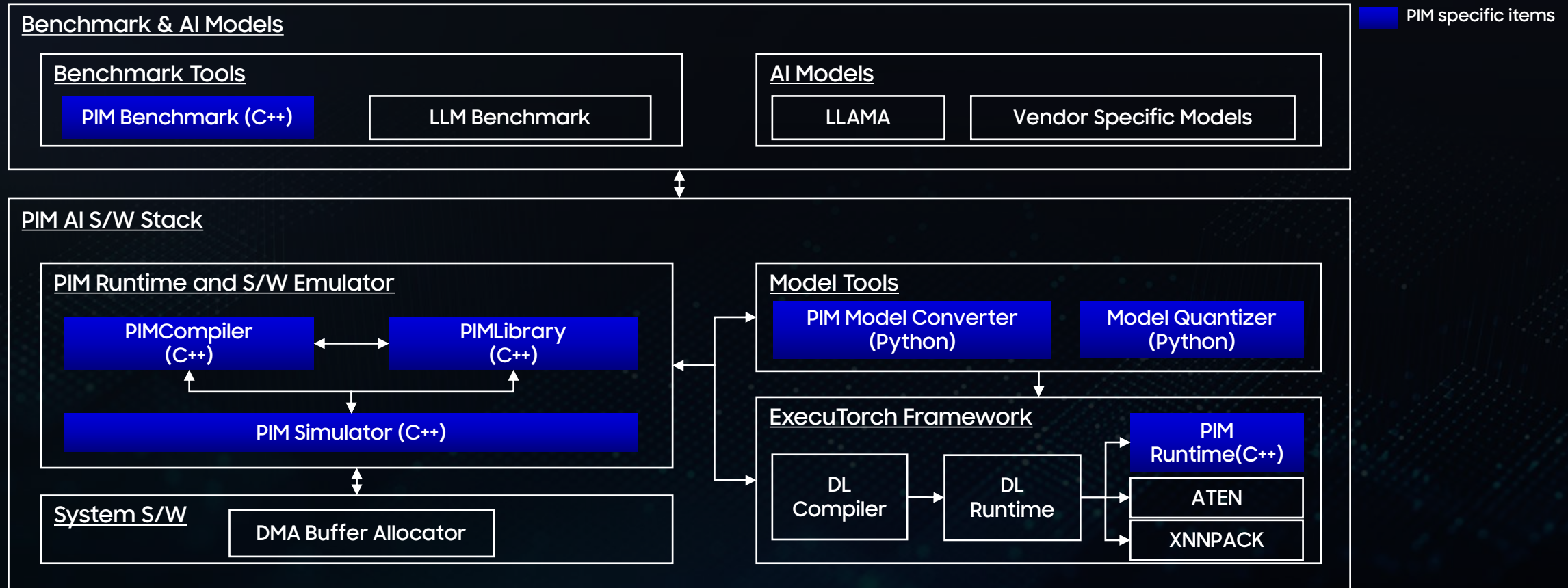
Appendix – Terms and Functions

- The exact function of the terms and functions are listed in the table below:

Term	Description
Source Register File (SRF)	Register used to buffer input activation vectors as input for a GEMV operation (input register)
Scale Register File (ScRF)	Register used to buffer scale factors to multiply with associated weight values for a GEMV operation
Vector Register File (VRF)	Register used to store partials sums during accumulation (output register)
Instruction Register File (IRF)	Register used to store 16b PIM instruction (total of 64 instructions can be stored)
Configuration Register File	Register used to configure various PIM block settings (ex : MAC precision datatype, bit-width)
Read PIM (RDP)	Reads a 32B value from the designated register in a single PIM block
Write PIM (WRP)	Writes a 32B value to the designated register in a single PIM block
Write PIM Broadcast (WRPB)	Writes a 32B value to the designated register in multiple PIM blocks
PIM Execution Read (PIMX-RD)	Performs a designated operation in all PIM blocks by reading values from DRAM
PIM Execution Write (PIMX-WR)	Writes the value of the designated register in each PIM block to DRAM

Appendix - LPDDR5X-PIM Software Development Kit

- PIM SDK that is used as reference to make PIM control software stack
 - Provides a toolchain for implementing PIM functions and evaluating performance
 - Contains LLM performance evaluation environment with *ExecuTorch framework



*ExecuTorch is an end-to-end solution for enabling on-device inference capabilities across mobile and edge devices including wearables, embedded devices and microcontrollers.

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